

Chapter 18

Carboxylic acids and derivatives

Balsamic vinegar with olive oil is a popular dip for bread in Mediterranean countries. The Mediterranean climate is suited to growing olive trees. Olives provide a ready source of olive oil when crushed, but making balsamic vinegar can be a twelve-year process.

Balsamic vinegar comes from the Emilia Romagna region of Italy and is made from grape juice. The grape juice is simmered to reduce the volume and the concentrate fermented to produce ethanol. Naturally occurring bacteria then oxidise the ethanol to ethanoic acid. Ethanoic acid gives vinegar its acidic taste and smell. But before balsamic vinegar is ready to use, the 'vinegar' is stored in wooden barrels for up to twelve years. This aging process gives the characteristic flavour of balsamic vinegar.

Prior understanding

You may recall some properties of acids from Chapter 7 and that ethanoic acid is a weak acid. You may remember the homologous series of carboxylic acids from Chapter 13. In Chapter 16, you read about the reduction reactions of carboxylic acids to produce a primary alcohol, and the oxidation reaction to produce a carboxylic acid from a primary alcohol. You may remember visiting these reactions again when reading about carbonyl compounds in Chapter 17.

Learning aims

In this chapter, you will find out about the properties of carboxylic acids. You will read about reactions that produce carboxylic acids, and reactions of carboxylic acids, including redox reactions, neutralisation, acid-base reactions, esterification and reduction reactions. You will revisit reactions that produce esters and find out about the hydrolysis of esters.

18.1 Carboxylic acids (Syllabus 13.1.3–13.1.5)

18.2 Reactions that produce carboxylic acids (Syllabus 18.1.1)

18.3 Reactions of carboxylic acids (Syllabus 18.1.2)

18.4 Esters (Syllabus 18.2.1 and 18.2.2)



Figure 18.1 Ethanedioic acid is a poisonous white crystalline solid. A molecule of ethanedioic acid contains two carboxylic acid functional groups. Salts of ethanedioic acid are found in rhubarb and are responsible for the sour taste.



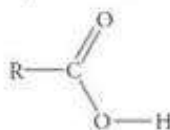
Figure 18.2 Methanoic acid is the simplest carboxylic acid. It used to be prepared by distilling red ants.



Figure 18.3 Benzoic acid is found in raspberries. A molecule of benzoic acid contains a carboxylic acid group attached to a benzene ring.

18.1 Carboxylic acids

Carboxylic acids are organic acids. They contain the -COOH functional group:



where R represents an alkyl group or an aryl group.

Many carboxylic acids occur naturally. Citric acid in fruit is responsible for its acidic taste. Methanoic acid is the acid in ant bites that causes the irritation. Benzoic acid occurs in raspberries. The lactic acid in milk products such as yogurt gives the sour taste.

Carboxylic acid names end in *-oic acid*. Table 18.1 gives the names and displayed formulae of the first four carboxylic acids.

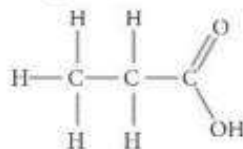
Carboxylic acid	Displayed formula
methanoic acid	$\begin{array}{c} \text{O} \\ \parallel \\ \text{H}-\text{C} \\ \\ \text{O}-\text{H} \end{array}$
ethanoic acid	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{C} \\ \quad \parallel \\ \text{H} \quad \text{O}-\text{H} \end{array}$
propanoic acid	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{C} \\ \quad \quad \parallel \\ \text{H} \quad \text{H} \quad \text{O}-\text{H} \end{array}$
butanoic acid	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C} \\ \quad \quad \quad \parallel \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{O}-\text{H} \end{array}$

Table 18.1 Carboxylic acids: names and displayed formulae.

Link

In Section 13.1, you found out how carboxylic acids are named.

Note that when naming carboxylic acids, the carbon atom in the carboxylic acid group counts as one of the carbons in the longest carbon chain. In this carboxylic acid:

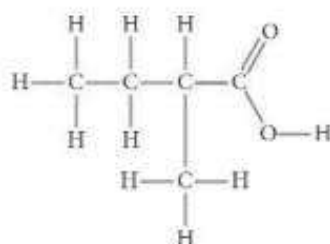


there are three carbon atoms, including the carbon atom in the functional group. The stem of the name is *propan*. The *-e* has been removed from propane to avoid a double vowel. This is propanoic acid.

Carboxylic acids can have side groups attached to the carbon chain. These form the suffix of the name, as in this worked example.

Worked example

Name this molecule:

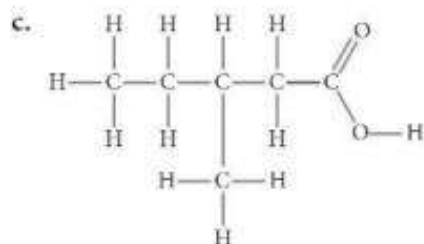
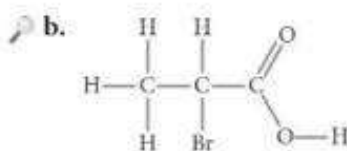


Answer

The -COOH group gives the suffix *-oic acid*. The longest carbon chain is four carbons long, which gives the name *butan-*. There is a methyl group on the second carbon atom. Note when numbering side groups in carboxylic acids, the carbon atom in the COOH group is always the first carbon atom. This methyl group gives the prefix *2-methyl*.

The name is 2-methylbutanoic acid.

1. Name these molecules:



2. A molecule of ethanedioic acid contains two carboxylic acid functional groups. Draw its displayed formula.

Tip

The old name for ethanoic acid is acetic acid. Acetic acid is commonly used on ingredient labels or in older chemistry books and some websites. Do not get confused.

BOILING POINTS

You have previously compared the boiling points of alkanes and alcohols. Alcohols have higher boiling points than the alkane with the same number of carbon atoms. For example, the boiling point of ethane is -89°C and the boiling point of ethanol is 78°C . The reason for ethanol's higher boiling point is the presence of hydrogen bonds between the ethanol molecules. These have to be broken so that ethanol can boil and that takes energy.

The boiling points of carboxylic acids are even higher than the boiling points of alcohols. Table 18.2 compares boiling points of alcohols with those of carboxylic acids.

Link

You can revisit Section 16.1 to compare the boiling points of alkanes and alcohols.

Alcohol	Boiling point of alcohol/ $^\circ\text{C}$	Carboxylic acid	Boiling point of carboxylic acid/ $^\circ\text{C}$
CH_3OH	65	HCOOH	101
$\text{CH}_3\text{CH}_2\text{OH}$	78	CH_3COOH	118
$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$	97	$\text{CH}_3\text{CH}_2\text{COOH}$	141
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$	100	$\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$	164

Table 18.2 Alcohol and carboxylic acid boiling points:

Link

You can refer back to Chapter 3 to read about hydrogen bonds and van der Waals' forces.

Note that propan-1-ol and ethanoic acid have the same relative molecular mass of 60, so the difference between their boiling points is not due to van der Waals' forces. The higher boiling points of carboxylic acids are due in part to hydrogen bonds.

Two molecules of a carboxylic acid are able to form two hydrogen bonds between them in such a way that the two molecules act as one. The hydrogen bonds form between the slightly negative oxygen atom of the C=O bond and the slightly positive hydrogen atom of the O-H bond. This double molecule is called a **dimer**. Figure 18.4 shows these hydrogen bonds between two molecules of ethanoic acid.

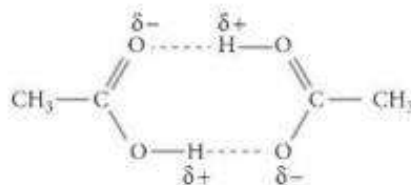


Figure 18.4 Ethanoic acid dimer.

Link

You can read Chapter 3 to revisit the link between relative molecular mass and van der Waals' forces.

The ethanoic acid dimer has a relative molecular mass of 120. This increases the van der Waals' forces between adjacent dimers and boiling points are higher.

SOLUBILITY

Carboxylic acids with up to four carbon atoms are soluble in water. These carboxylic acids do not form dimers when mixed with water, but form hydrogen bonds with water molecules instead. As the alkyl groups in the carboxylic acids get larger, solubility decreases because the alkyl groups cannot form hydrogen bonds with water. The solubility of carboxylic acids is a balance between the parts of the molecule that can form hydrogen bonds with water and the alkyl groups, which cannot.

When a carboxylic acid such as ethanoic acid dissolves in water, another reaction occurs. Ethanoic acid ionises to produce the hydrogen ions responsible for its acidic behaviour.

3. Draw a diagram to show a hydrogen bond between a water molecule and a molecule of ethanoic acid.
- ★ 4. Alcohols, aldehydes, ketones and carboxylic acids have similar solubility trends in water.
 - a. Describe the trends.
 - b. Explain why the trends are similar.

Link

You can revisit Chapter 7 to read about acids. Later in this chapter, you will read about the acidic properties of carboxylic acids.

Key ideas

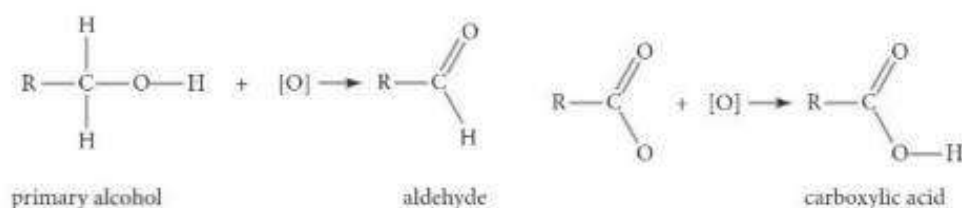
- Carboxylic acids have the functional group -COOH .
- The names of carboxylic acids end in *-oic acid*.
- Hydrogen bonds between two carboxylic acid molecules produce a double molecule called a dimer.
- Carboxylic acids have higher boiling points than alcohols because of the presence of dimers.
- Carboxylic acids with up to four carbon atoms in their molecules are soluble in water.
- The solubility of carboxylic acids decreases when there are more than four carbon atoms in their molecules.

18.2 Reactions that produce carboxylic acids

THE OXIDATION OF A PRIMARY ALCOHOL OR ALDEHYDE

Primary alcohols and aldehydes can be oxidised to produce a carboxylic acid. Suitable oxidising agents are acidified potassium dichromate(VI) solution or acidified potassium manganate(VII) solution. The reaction takes place under reflux.

The reaction to oxidise primary alcohols takes place in two stages:



A similar reaction takes place in the body when alcohol is consumed. Ethanol is the alcohol in alcoholic drinks. Firstly, ethanol is broken down in the liver to ethanal by an enzyme. Ethanal is the cause of headaches and nausea, so needs to be broken down rapidly. Another enzyme oxidises the ethanal to ethanoic acid. Ethanoic acid can then be broken down into carbon dioxide and water and excreted from the body.

5. Explain why secondary and tertiary alcohols do not readily oxidise to carboxylic acids.
6. The oxidation of ethanol to ethanoic acid takes place under reflux. If the reaction mixture is distilled as soon as the reactants are mixed, what is the product?
7. Name the organic product of these reactions:
 - a. propan-1-ol is reacted with acidified potassium dichromate(VI) and the reaction mixture immediately distilled
 - b. propan-1-ol is reacted with acidified potassium dichromate(VI) and the reaction mixture refluxed for 30 minutes
 - c. propan-2-ol is reacted with acidified potassium manganate(VII) and the reaction mixture distilled.

Link

Section 16.3 covers the oxidation of alcohols to produce aldehydes, ketones and carboxylic acids. The same reactions are also in Section 17.2 as methods to prepare aldehydes and ketones.

Experimental skills 18.1: Oxidising ethanol to produce ethanoic acid

Link

The oxidation of ethanol to ethanal is covered in Section 17.1.

In Chapter 17, you found out how ethanol can be oxidised to ethanal in the laboratory. This is a similar oxidation reaction, but the reaction conditions are changed so that ethanoic acid is produced.

Apparatus

You will need the following apparatus:

- a pear-shaped flask
- a condenser
- a heating mantle
- an ice bath
- a clamp and stand
- anti-bumping granules
- ethanol
- acidified potassium manganate(VII) solutions.

Technique

The acidified potassium manganate(VII) is placed in the pear-shaped flask and cooled in an ice bath. A few anti-bumping granules are added and the condenser connected to the flask. The water is turned on so that a steady stream flows through the condenser.

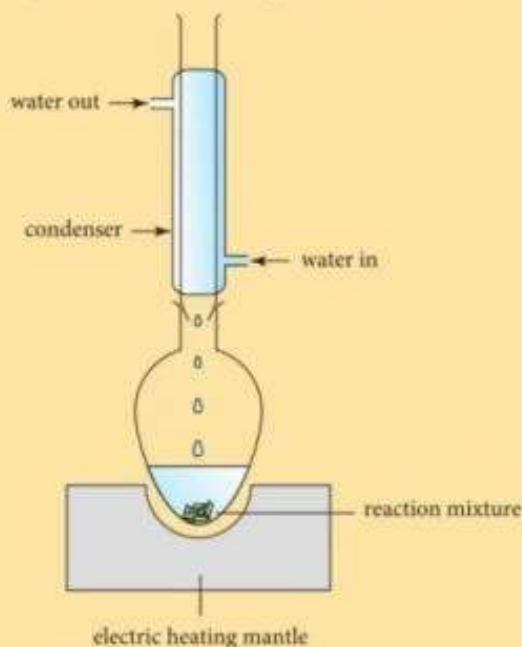


Figure 18.5 Refluxing apparatus.

Ethanol is added, a few drops at a time, down the condenser. It is important that the ethanol is added slowly and the reaction mixture gently mixed to avoid

a vigorous reaction. The reaction is exothermic. When the reaction has subsided, the ice bath is removed and the outside of the flask dried thoroughly. The flask is placed in the heating mantle and the reaction mixture heated. Steady boiling needs to be maintained for about 30 minutes.

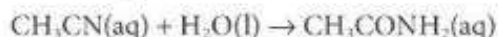
The flask should contain ethanoic acid at the end of this time.

- P1. Why is it necessary to cool the oxidising agent and connect the condenser before the alcohol is added?
- P2. Why must the oxidising agent be in excess in the reaction mixture?
- P3. Which is the correct answer?
- Potassium manganate(VII) solution is purple. During the reaction, the colour changes from purple to colourless. The potassium manganate(VII) solution is:
- | | |
|------------|-------------|
| A used up | C reduced |
| B oxidised | D condensed |
- P4. What method can be used to separate the ethanoic acid product from the mixture in the flask?
- P5. Ethanoic acid can be identified from its vinegary smell. How would you safely smell a sample of ethanoic acid?

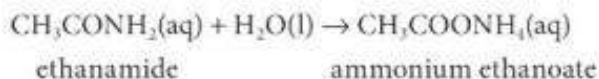
THE HYDROLYSIS OF A NITRILE

Nitriles have the functional group -CN .

Nitriles react with water in two stages. In the reaction between ethanenitrile and water, an amide is produced in the first stage of the reaction:



The amide is then hydrolysed to ammonium ethanoate



Ammonium ethanoate is a salt of ethanoic acid.

This is a hydrolysis reaction. The problem is that this reaction is very slow. The reaction has a faster rate if the nitrile is heated under reflux with either a dilute acid, such as hydrochloric acid or sulfuric acid, or a dilute alkali, such as sodium hydroxide. Both dilute acid and dilute alkali give the same products in the end, but the reaction sequences are different.

If ethanenitrile is heated under reflux with dilute hydrochloric acid, ammonium ethanoate is produced as in the hydrolysis reaction above. Ammonium ethanoate then reacts with dilute hydrochloric acid to produce ethanoic acid and ammonium chloride:

**Link**

You read about the homologous series of nitriles in Section 13.1. You will find out more about nitriles in Chapter 19.

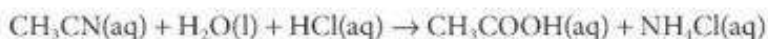
Link

You will find out more about salts of ethanoic acid later in this chapter, in Section 18.3.

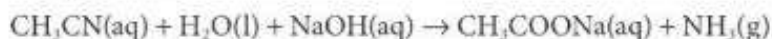
Ethanoic acid is a weak acid and is formed in this reaction because once the ethanoate ions and hydrogen ions collide and react, ethanoic acid molecules tend to remain.

The ethanoic acid can be separated by distillation.

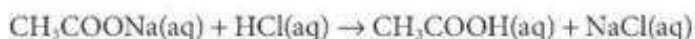
The overall reaction is:



If ethanenitrile is heated under reflux with dilute sodium hydroxide, any ethanoic acid formed reacts with sodium hydroxide to produce the salt. Sodium ethanoate and ammonia are produced:



In order to produce ethanoic acid, dilute hydrochloric or sulfuric acid needs to be added. If dilute hydrochloric acid is added to the reaction mixture:



The ethanoic acid can be separated by distillation.

THE HYDROLYSIS OF AN ESTER

Esters are formed in reactions between an alcohol and a carboxylic acid. Esters have the general formula:



where R is the part of the ester molecule from the carboxylic acid and R¹ is the part of the ester molecule from the alcohol. When ethanoic acid reacts with ethanol, the ester ethyl ethanoate is produced.

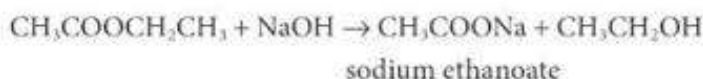
The reaction can be reversed in a hydrolysis reaction. Although this is essentially a reaction with water, the reaction is very slow if water alone is used. The rate increases if dilute acid or alkali is used as a catalyst. The reaction takes place when the reactants are heated under reflux.

Ethyl ethanoate can be hydrolysed to produce ethanoic acid and ethanol using dilute hydrochloric acid. The reaction is:

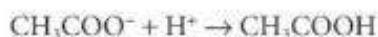


Since the reaction is reversible, an excess of water is needed to ensure the best yield of ethanoic acid and ethanol. The water comes from the dilute acid and the dilute acid must be in excess.

The hydrolysis reaction can also be carried out using a dilute alkali like dilute sodium hydroxide. This reaction takes place when heated under reflux. In this case, sodium ethanoate is produced:



The advantage of this reaction is that it is not reversible and the products are relatively easy to separate. The ethanol can be distilled off. A strong acid like dilute hydrochloric acid can be added to the remaining solution after distillation. The hydrogen ions react with the ethanoate ions as follows:



Since ethanoic acid is a weak acid, the ions tend to stay combined.

Link

You found out about strong and weak acids in Chapter 7.

Link

In Section 13.1, you read about names of esters.

Link

You found out how esters are formed from alcohols and carboxylic acids in Section 16.3 and how they can be hydrolysed in Section 16.2.

Link

You can revisit chemical equilibria in Chapter 7.

18 Carboxylic acids and derivatives

8. Name the carboxylic produced when propanenitrile is hydrolysed using dilute hydrochloric acid.
9. During the hydrolysis of ethanenitrile, ethanoate ions react with hydrogen ions to produce ethanoic acid. Write an equation to show the reaction.
10. Name the carboxylic acid produced when ethyl propanoate is hydrolysed.
11. The hydrolysis of ethyl ethanoate to produce ethanoic acid and ethanol is a reversible reaction:



The reaction is catalysed by hydrogen ions.

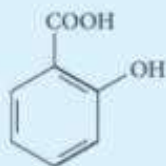
What happens to the position of equilibrium if:

- a. a concentrated strong acid is added to the mixture
- b. a dilute strong acid is added to the mixture?

Assignment 18.1: The hydrolysis of aspirin

Aspirin is a common medicine used world-wide to treat pain and other conditions.

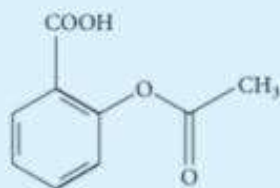
Extracts from willow bark have been used to treat pain for thousands of years. The active ingredient in willow bark is an organic acid called 2-hydroxybenzoic acid. Its old name, salicylic acid, is still commonly used. The structure of 2-hydroxybenzoic acid is:



It was effective against pain, but also had many side effects. During the nineteenth century, drug companies experimented to find an alternative with fewer side effects.

The result was 2-ethanoyloxybenzenecarboxylic acid. Its old name, acetyl salicylic acid, is also still commonly used, but using the name aspirin is much simpler.

This is its structure:



This is the active ingredient in aspirin tablets that we use today.

- A1. Copy the structural formula of aspirin and circle and label the following groups:
 - a. a carboxylic acid group
 - b. an ester group.
- A2. Aspirin can be prepared by reacting 2-hydroxybenzoic acid with a carboxylic acid. Name the carboxylic acid.
- A3. Aspirin has a shelf life of two to three years. After that, you may find a vinegary smell when you open an old bottle of aspirin. A chemical reaction has taken place.
 - a. What type of reaction has occurred?
 - b. Write an equation to show this reaction using structural formulae.
 - c. Why might this reaction happen more quickly if you store aspirin in a bathroom?
 - d. This reaction can be carried out in the laboratory. What reaction conditions are needed?
- A4. Soluble aspirin is the sodium salt of aspirin. Draw its structural formula.



Figure 18.6 Aspirin tablets are produced by the pharmaceutical industry. The forerunners to aspirin were extracted from natural sources like willow bark.

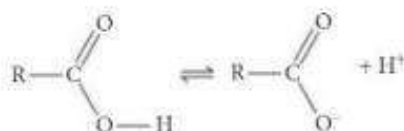
Key ideas

Reactions that produce carboxylic acids are:

- the oxidation of primary alcohols using acidified potassium dichromate(VI) or acidified potassium manganate(VII)
- the oxidation of aldehydes using acidified potassium dichromate(VI) or acidified potassium manganate(VII)
- the hydrolysis of nitriles using dilute acid or dilute alkali followed by dilute acid
- the hydrolysis of an ester using dilute acid or alkali followed by dilute acid.

18.3 Reactions of carboxylic acids

Carboxylic acids are acidic because the hydrogen atom in the OH bond can produce hydrogen ions:



Dilute aqueous hydrochloric acid is completely ionised. Carboxylic acids are only partially ionised. This makes them weak acids. Depending on concentration, roughly, no more than 1% of a carboxylic acid is ionised: 99% of a solution of ethanoic acid exists as molecules.

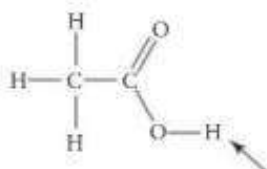
Like all acids, the hydrogen in a carboxylic acid can be replaced with a metal atom to produce a salt. Salts of carboxylic acids have names ending *-oate*. The number of carbon atoms gives the stem of the name, with the 'e' removed, and the metal gives the first part of the name.

Worked example

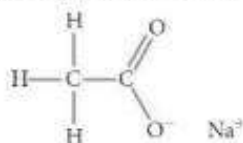
Draw the displayed formula and name the salt produced when a sodium atom replaces the hydrogen atom in ethanoic acid.

Answer

Ethanoic acid is:



The arrow points to the hydrogen atom that is replaced by a sodium atom. Note that the type of bond changes. The O–H bond is covalent and we use a single line to represent it. The bond between the oxygen and sodium is ionic. This is written as:



This is the displayed formula. This compound is an ionic salt and is named as such.

Link

In Section 16.5, you read why ethanoic acid is acidic.

Link

In Chapter 7, you found out about strong and weak acids.

Tip

The sodium ion has one positive charge and the ethanoate ion has one negative charge. Remember that some metals can form ions with more than one positive charge.

Tip

In the worked example, the formula can be written as CH_3COONa or $\text{CH}_3\text{COO}^- \text{Na}^+$.

The first part of the name is *sodium*. The stem of the name is *ethan* since there are two carbon atoms and the name ends in *oate*.

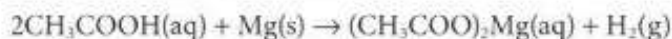
The name is sodium ethanoate.

12. Write the structural formula for the salt formed when calcium replaces hydrogen in ethanoic acid.
13. Laboratory solutions of ethanoic acid have a pH between 2 and 3. Suggest why a solution of methanoic acid with the same concentration has a pH 0.5 units lower.

THE REACTIONS OF CARBOXYLIC ACIDS WITH METALS

You should remember from your previous courses that acids react with metals to produce a salt and hydrogen. Carboxylic acids have similar reactions and react with reactive metals to produce a salt and hydrogen.

Magnesium reacts with dilute ethanoic acid to give off hydrogen gas and produce a solution of magnesium ethanoate. The reaction is:



You can compare this with the reaction between dilute hydrochloric acid of the same concentration and magnesium:



Both reactions **effervesce**, but the reaction between ethanoic acid and magnesium is much slower because the concentration of hydrogen ions in ethanoic acid is lower.

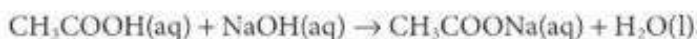
Reactions between reactive metals and carboxylic acids are redox reactions because:

- The metal loses electrons to form a positive ion. The metal is oxidised.
- The COO^- group gains an electron to form a negative ion. The COO^- group is reduced.

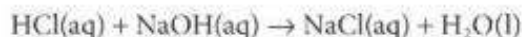
THE REACTIONS OF CARBOXYLIC ACIDS WITH ALKALIS

You may also remember from your previous course that acids and alkalis react in neutralisation reactions to produce a salt and water only. Carboxylic acids also have similar reactions and react with alkalis to form a salt and water.

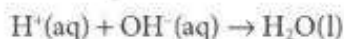
If sodium hydroxide solution is added to dilute ethanoic acid, a solution of sodium ethanoate is produced. The reaction is:



You can compare this with the reaction between sodium hydroxide solution and dilute hydrochloric acid:



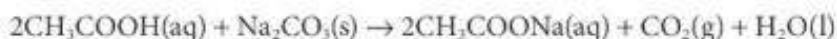
Both reactions can be represented by:



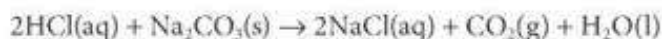
THE REACTIONS OF CARBOXYLIC ACIDS WITH CARBONATES

Carboxylic acids react with carbonates to produce a salt, carbon dioxide and water. Acids such as hydrochloric acid have similar reactions.

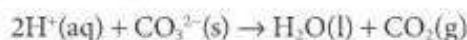
If sodium carbonate is added to dilute ethanoic acid, there is an immediate effervescence as carbon dioxide is given off and a solution of sodium ethanoate is produced:



Again, you can compare this with the reaction between dilute hydrochloric acid and sodium carbonate:



Both reactions can be represented by:



14. Write full equations, including state symbols, for the following reactions:
- dilute ethanoic acid and calcium metal
 - dilute propanoic acid and copper carbonate
 - dilute propanoic acid and potassium hydroxide solution
 - dilute methanoic acid and sodium metal
 - dilute butanoic acid and calcium carbonate.
15. a. Compare the rate of reaction when 0.5 g sodium carbonate is added to 20 cm³ of 1.00 mol dm⁻³ ethanoic acid and to 20 cm³ of 1.00 mol dm⁻³ hydrochloric acid.
b. Explain your answer to a.

ESTERIFICATION

Carboxylic acids react with alcohols in the presence of an acid catalyst to produce esters. The catalyst is usually concentrated sulfuric acid. You will read more about esterification in the next section.

REDUCTION OF CARBOXYLIC ACIDS

Primary alcohols are produced when carboxylic acids are reduced. Lithium tetrahydridoaluminate, LiAlH₄, is a suitable reducing agent.

The carboxylic acid is firstly reduced to an aldehyde, then the aldehyde is reduced to an alcohol.

In this reaction:



Ethanoic acid is reduced to ethanal then to ethanol.

Key ideas

Reactions of carboxylic acids include:

- the redox reaction with metals to produce a salt and water
- the neutralisation reaction with alkalis to produce a salt and water
- the acid-base reaction with carbonates to produce a salt, carbon dioxide and water
- esterification with alcohols
- reduction using lithium tetrahydridoaluminate.

Link

In Chapter 16, you read about reactions to produce esters and how esters are named. Earlier in this chapter you found out how esters are hydrolysed.

Link

In Section 16.2, you read about the reduction of carboxylic acids using LiAlH₄ as the reducing agent.

18.4 Esters

FORMATION OF ESTERS

Esters are formed in condensation reactions between a carboxylic acid and an alcohol. The reaction is also referred to as **esterification**. You have read about esters earlier in this book. You can revise them by working through Assignment 18.2.

Link

Names, formulae and the preparation of esters are covered in more detail in Section 13.1 and Section 16.3.

Link

The hydrolysis of an ester as a means of producing a carboxylic acid was covered earlier in this chapter.

HYDROLYSIS OF ESTERS

The reverse reaction of esterification is the hydrolysis of an ester to produce a carboxylic acid and an alcohol. You have already read about this in this chapter.

Assignment 18.2: Ester smells

Different carboxylic acids and alcohols produce different esters with different smells. Table 18.3 gives the names of the carboxylic acid and alcohol used to make the ester and the smell of the ester.

Carboxylic acid	Alcohol				
	methanol	ethanol	propan-1-ol	butan-1-ol	pentan-1-ol
methanoic acid	cleaning fluid	cleaning fluid	apples	raspberries	fruity
ethanoic acid	cleaning fluid	glue	pears	apples	bananas
propanoic acid	fruity	pineapple	fruity	apple	apricots
butanoic acid	pineapple	pineapple	pears	pineapple	apricots
pentanoic acid	flowers	fruity	pineapple	fruity	apples

Table 18.3 Names of carboxylic acid and alcohol used to make an ester and the smell of the ester.

- A1. Name and give the structural formula for the esters responsible for the:
 - a. raspberry smell
 - b. banana smell.
- A2. Outline the stages to prepare a sample of the ester responsible for the banana smell.
- A3. Butanoic acid is the acid found in rancid butter, Parmesan cheese and vomit. It has a disgusting smell. How could you make sure that esters made from butanoic acid did not also smell of butanoic acid? Explain your answer.
- A4. Esters that are used for their pleasant aromas need to be volatile so that they evaporate easily and molecules can be detected by our noses. Table 18.4 gives the boiling points of ethyl ethanoate and a carboxylic acid with the same relative molecular mass, butanoic acid:

Compound	Relative molecular mass	Boiling point/°C
ethyl ethanoate	88	77
butanoic acid	88	164

Table 18.4

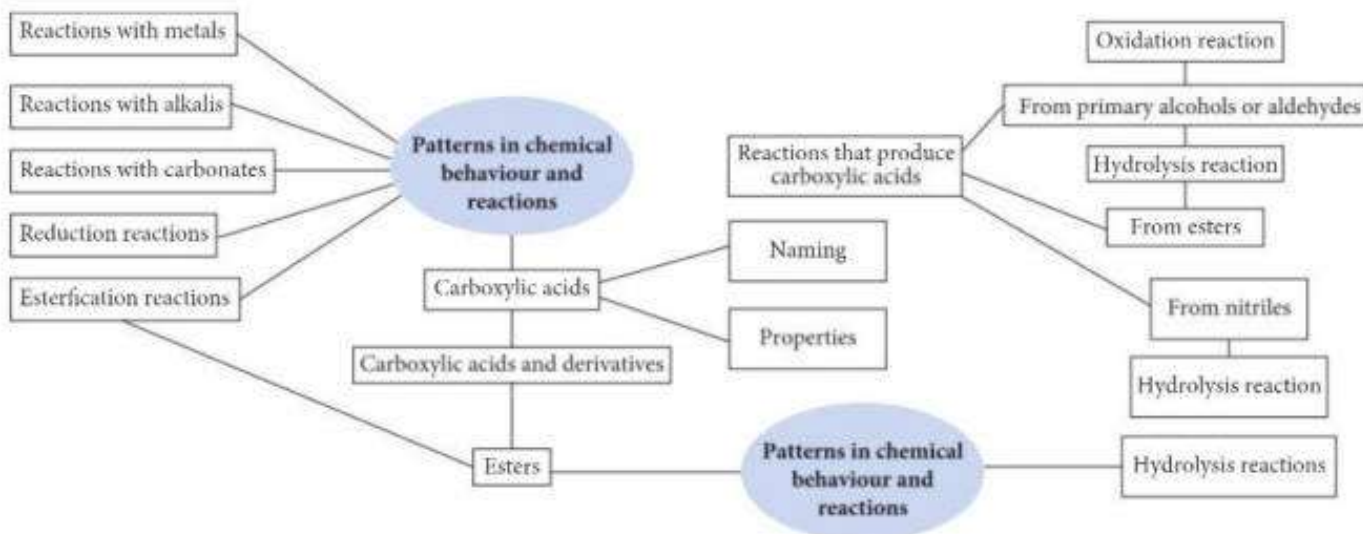
Suggest why the boiling point of ethyl ethanoate is much lower than the boiling point of butanoic acid.



Figure 18.7 Many of the smells from these fruits are esters.

CHAPTER OVERVIEW

Try copying this mini mind map and expanding it. Use your notes from other chapters to help you explore how the essential ideas, theories and principles can be linked further together.



WHAT YOU HAVE LEARNED

Now that you have finished this chapter you should be able to:

- recall that the oxidation of primary alcohols and aldehydes produce carboxylic acids
- recall that the hydrolysis of nitriles produces carboxylic acids
- recall that the hydrolysis of esters produces carboxylic acids
- describe the redox reaction between carboxylic acids and metals
- describe the neutralisation reaction between carboxylic acids and alkalis
- describe the acid–base reaction between carboxylic acids and carbonates
- describe esterification with alcohols with concentrated sulfuric acid as a catalyst
- describe reduction reactions of carboxylic acids
- recall condensation reactions to produce esters
- describe the hydrolysis of esters.

CHAPTER REVIEW

1. What is the name given to the 'double molecule' formed by two carboxylic acid molecules with two hydrogen bonds between them?
2. Esters can be hydrolysed by refluxing with dilute acid or alkali. Since this is a hydrolysis reaction, what role does the dilute acid or alkali play?
3. Name the salt formed when magnesium reacts with dilute ethanoic acid and write its structural formula.
4. Write a full equation for the reaction between butanoic acid and magnesium.
5. Write a full equation for the reaction between butanoic acid and magnesium carbonate.
6. Write a full equation for the reduction of propanoic acid to produce a primary alcohol.
7. Why is the reaction between an alcohol and a carboxylic acid to produce an ester called a condensation reaction?
8. What is the ester functional group?
9. Name the ester formed when propan-1-ol reacts with pentanoic acid and write its structural formula.
10. An ester can be hydrolysed by refluxing with dilute sodium hydroxide solution. Why is it necessary to add dilute acid to the product to produce a carboxylic acid?

Chapter 33

Carboxylic acids and derivatives

If you have ever been bitten by an ant, then you will know that it is painful and can lead to swelling of the bitten area. In an ant's bite there is a carboxylic acid called methanoic acid, HCOOH . The traditional name for this acid is formic acid because it was first extracted from ants and the genus name for ant is *Formica*. A household remedy for an ant bite is a paste made of water and baking soda. Why baking soda? Baking soda contains sodium hydrogencarbonate, which is an alkali. As an alkali, baking soda will neutralise the acid present and provide some relief from the bite. In this chapter we will look at organic acids, such as methanoic acid, and a wide range of related compounds.

Prior understanding

From your previous chemistry course you may recall that ethanoic acid (which is found in vinegar) is a carboxylic acid and is made by the oxidation of ethanol. From your previous chemistry course at AS, you may remember that primary alcohols and aldehydes can be oxidised to form carboxylic acids. In addition, the hydrolysis of esters or nitriles is an alternative way of making carboxylic acids. You may remember that carboxylic acids are described as weak acids, but that they do still react as typical acids in their reactions with alkalis, carbonates and metals. Finally, your AS chemistry course looked at how esters can be produced from the reaction of organic acids with alcohols.

Learning aims

In this chapter you will learn about some of the reactions of carboxylic acids and how they can be further oxidised. You will also learn how to compare (i) the relative acidities of carboxylic acids, phenols and alcohols, (ii) the relative acidities of **chlorine-substituted carboxylic acids**. Further, you will consider the hydrolysis of **acyl chlorides**, including the **addition-elimination** reaction mechanism, and you will learn how acyl chlorides react to produce esters and **amides**.

33.1 Production of benzoic acid (Syllabus 33.1.1)

33.2 Reactions of carboxylic acids (Syllabus 33.1.2, 33.3.1)

33.3 Further oxidation of carboxylic acids (Syllabus 33.1.3)

33.4 Acidity of carboxylic acids (Syllabus 33.1.4)

33.5 Acidity of chlorine-substituted carboxylic acids (Syllabus 33.1.5)

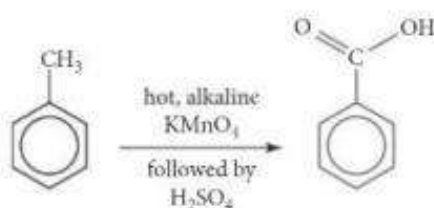
33.6 Reactions of acyl chlorides (Syllabus 33.2.1–33.3.4)

33.1 Production of benzoic acid

OXIDATION OF ALKYL BENZENES

When alkylbenzenes are heated with alkaline potassium manganate(VII) and the reaction mixture is acidified with dilute acid, benzoic acid is produced.

This formation of benzoic acid from methylbenzene can be represented by:



Note that this has been represented by a type of flow diagram rather than a balanced equation. The situation is too complicated to represent in one simple equation so don't try. This difficulty is evident in the fact that other alkylbenzenes, such as ethylbenzene, propylbenzene etc. are also oxidised under these conditions to benzoic acid.

33.2 Reactions of carboxylic acids

REACTIONS WITH PHOSPHORUS CHLORIDES

Carboxylic acids react with phosphorus chlorides (PCl_3 and PCl_5) to produce acyl chlorides. In these reactions the —OH group is replaced, just as it would be in alcohols.

For example, the reaction of ethanoic acid with phosphorus(III) chloride, PCl_3 , is:



The reaction of ethanoic acid with phosphorus(V) chloride, PCl_5 , is:



Worked example

Write the equation for the formation of propanoyl chloride using phosphorus(III) chloride. State symbols are not required.

Answer

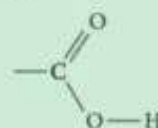
Propanoyl chloride has three carbon atoms and so has the structural formula $\text{CH}_3\text{CH}_2\text{COCl}$. Propanoyl chloride will be formed from propanoic acid. The equation will be very similar to that shown previously with ethanoic acid:

Link

You learned about the oxidation of alkenes by KMnO_4 in Chapter 14.

Tip

Carboxylic acids contain the —COOH group:



Acyl chlorides contain the —COCl group:



Link

You learned about the reaction of alcohols with PCl_3 and PCl_5 in Chapter 16.

Tip

Make sure that the correct phosphorus chloride compound is referred to because different products result. The key thing to focus on is the oxidation number stated after the name.



The question does not state a particular type of formula. So any appropriate type such as displayed or skeletal can be used. A molecular formula would not be appropriate because it does not specifically identify the molecule propanoyl chloride—other compounds have the same molecular formula.

Link

You may remember that oxidation numbers were considered in Chapter 2 and are an indication of the combining power of the atom.

Link

Look back at Chapter 2 to revise how to work out reacting masses and volumes of solutions and gases.

Link

Chapter 16 covers the oxidation of primary alcohols to carboxylic acids.

Link

You may remember that in oxidation reactions of alcohols and aldehydes that the oxidising agent is represented simply as [O].

Link

Chapter 17 looks at the use of Fehling's reagent and Tollens' reagent as tests for aldehydes. These are oxidation reactions.

REACTION WITH SOCl₂

Carboxylic acids also react with sulfur dichloride oxide, SOCl₂, to produce acyl chlorides.

For example:



The two by-products are gases, so at first it appears that the acyl chloride product is easier to obtain, but it will still have to be separated from any excess reactant.

- Look at the reactions between a carboxylic acid and:
 - phosphorus chlorides
 - sulfur dichloride oxide.
 Describe two similarities and three differences in these reactions.
- How would the acyl chloride be separated from the reaction mixture in the reactions between a carboxylic acid and:
 - phosphorus chlorides
 - sulfur dichloride oxide.
- Write the equation for the complete reaction of ethanedioic acid, HOOC-COOH, with phosphorus(III) chloride.
- Write the equation for the reaction of ethanedioic acid with sulfur dichloride oxide.
- Calculate the volume of hydrogen chloride gas given off in the complete reaction of 6.00 g of ethanoic acid with excess phosphorus(V) chloride. (Assume 1 mole of any gas occupies 24 dm³.)

Key idea

→ Carboxylic acids react with PCl₃, PCl₅ and SOCl₂ to produce acyl chlorides

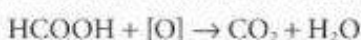
33.3 Further oxidation of carboxylic acids

The oxidation of primary alcohols produces carboxylic acids. For example, propan-1-ol is oxidised to propanoic acid:



When acidified potassium dichromate(VI) is used, the oxidation reaction usually stops with the carboxylic acid.

Methanoic acid is an exception. Unlike other carboxylic acids, it can be further oxidised using acidified potassium dichromate(VI) to produce carbon dioxide and water. This is because it can be oxidised further without the need to break a C—C bond:



Methanoic acid can react with Fehling's reagent and also with Tollens' reagent. It is further oxidised with both reagents to form carbon dioxide and water.

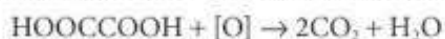
There is one more oxidising agent that will further oxidise methanoic acid.

Methanoic acid is oxidised by potassium manganate(VII) and the products are carbon dioxide and water. This reaction can use either a cold dilute solution or a hot concentrated solution of acidified potassium manganate(VII).

There is one more exception that you need to be aware of. Ethanedioic acid is a dicarboxylic acid. Its structural formula is HOOC-COOH and its displayed formula is:



This particular dicarboxylic acid is oxidised by warm acidified potassium manganate(VII) to produce carbon dioxide and water. The reaction is:



Link

Chapter 14 looks at the oxidation of alkenes using acidified potassium manganate(VII), KMnO_4 .

Tip

Other dicarboxylic acids react differently. Ethanedioic acid is a special case, so try to remember that.

Experimental skills 33.1: Determination of KMnO_4 concentration

A redox titration using a standard solution of oxalic acid can be used to determine the unknown concentration of a solution of potassium manganate(VII).

Apparatus

You will need the following apparatus:

- burette
- pipette and filler
- hot plate
- funnel
- oxalic acid, $\text{H}_2\text{C}_2\text{O}_4$
- KMnO_4 solution
- white tile
- deionised water
- beaker
- glass rod
- two decimal point balance
- conical flasks
- sulfuric acid, 1.0 mol dm^{-3}
- thermometer.

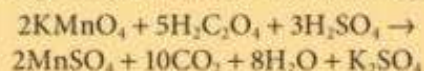
Technique

You will need access to a water supply, a sink and electricity.

A known mass of oxalic acid is weighed out and transferred to a volumetric flask with deionised water to the line in order to make a standard solution. A pipette is used to measure 25.0 cm^3 volumes in separate conical flasks. The solution is further acidified with sulfuric acid. The burette is filled with the

potassium manganate(VII) solution. The contents of the conical flask are heated to about 60°C , then removed from the hot plate. Then the potassium manganate(VII) solution is added from the burette.

The average titre is recorded. The overall equation is:



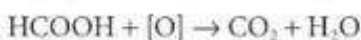
- P1. Calculate the concentration of the oxalic acid solution if 3.15 g of oxalic acid is used to produce a 250.0 cm^3 solution.
- P2. What is the oxidation number change of the carbon in this reaction. So what is the role of oxalic acid in this reaction?
- P3. Why is sulfuric acid needed for the titration when oxalic acid is itself an acid?
- P4. Suggest why the oxalic acid is heated before the titration. Give a possible reason for what might happen if the heating goes above 60°C .
- P5. What is the colour change at the end-point of the titration?
- P6. Calculate the concentration of the potassium manganate(VII) solution given an oxalic acid solution of $0.150 \text{ mol dm}^{-3}$ and an average titre of 12.10 cm^3 .

Worked example

Write the equation for the reaction between Fehling's reagent and methanoic acid.

Answer

Fehling's reagent, like all oxidising agents, is represented by [O] in an equation, so:



6. Write the equation for the complete oxidation of methanol using acidified potassium dichromate(VII), $\text{K}_2\text{Cr}_2\text{O}_7$, to produce carbon dioxide and water.
7. Describe the change in the appearance of the reaction mixture when methanoic acid is added to Fehling's reagent and when it is added to Tollens' reagent.
8. In a positive result for the use of Tollens' reagent the silver ions are reduced. Which of the following equations is correct to represent this reduction?

A $\text{Ag} \rightarrow \text{Ag}^+ + \text{e}^-$	C $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$
B $\text{Ag}^+ \rightarrow \text{Ag}^{2+} + \text{e}^-$	D $2\text{Ag}^+ + 2\text{e}^- \rightarrow \text{Ag}_2$
9. Compare the reaction of cold, dilute potassium manganate(VII) with methanoic acid to the reaction of that reagent with ethene.
- ★ 10. Write the overall equation for the reaction of 2-hydroxyethanoic acid with warm acidified potassium manganate(VII).

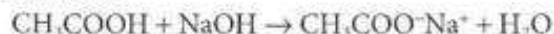
Key ideas

- Most carboxylic acids cannot be further oxidised.
- Methanoic acid can be further oxidised to carbon dioxide and water.
- Ethanedioic acid can be further oxidised to carbon dioxide and water.

33.4 Acidity of carboxylic acids

CARBOXYLIC ACIDS REACTING AS ACIDS

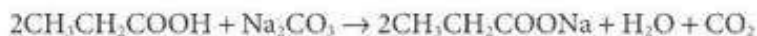
Carboxylic acids also react with sodium hydroxide. For example, the reaction of ethanoic acid with sodium hydroxide is:



The introduction to this chapter highlighted the reaction between a carboxylic acid, such as methanoic acid, and sodium hydrogencarbonate. Neither alcohols nor phenols react with carbonates, but carboxylic acids do react.

A different carboxylic acid, such as propanoic acid, can be used to show the reaction:

- with sodium carbonate



- with sodium hydrogen carbonate



Both sodium carbonate and sodium hydrogen carbonate are soluble in water. If either of these solutions is added to a carboxylic acid, then evidence of a reaction is the observation of fizzing or bubbles.

If asked to describe this reaction, all you can say is that you can see bubbles. The observation of fizzing does not prove that the gas is carbon dioxide.

Link

From Chapter 32 you may remember that phenol has stronger acidic properties than ethanol. Phenol reacts with sodium hydroxide, but ethanol does not.

Tip

The test for carbon dioxide is to bubble the gas through limewater, calcium hydroxide solution, and to look for the formation of a white precipitate.

COMPARING ACIDITY

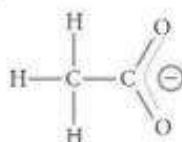
When considering the reason for the greater acidity of carboxylic acids over phenols, it is helpful to draw out the ionisation of a carboxylic acid to show the bonds involved. The equation for the ionisation of ethanoic acid can be drawn as follows:



The negative charge of the carboxylate ion can actually be on either of the two oxygen atoms. The following diagram shows how electron pairs can move to achieve this:



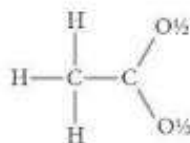
This shows that the negative charge on the carboxylate ion can be written on either of the two oxygen atoms. Studies of the carboxylate ion have shown that there is actually delocalisation of the electron density as illustrated below:



The evidence for this delocalisation of electron density is that the two C—O bonds in the carboxylate ion are equal in length and intermediate in strength over the O—C—O area. This provides stabilisation of the anion. So it causes the carboxylic acid to be a stronger acid than both phenol and ethanol.

Worked example

A student wrote the ionised form of ethanoic acid as shown. Explain why the student has done this.



Answer

The negative charge on the CH_3COO^- anion is distributed between the two oxygen atoms. So the student has attempted to illustrate this through the use of two half charges.

Link

Chapter 7 looks at acids and bases. Ethanoic acid is a weak acid and is only partially dissociated in aqueous solution. Strong acids, such as hydrochloric acid, are completely dissociated.

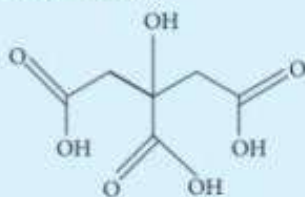
- Write the equation for the reaction of ethanedioic acid with sodium hydrogencarbonate.
- Draw a labelled diagram of the apparatus that could be used to confirm that a liquid is a carboxylic acid from its reaction with sodium carbonate.
- Calculate the mass of sodium carbonate that would be required to completely react with 3.70 g of propanoic acid.
- Suggest a bond length for the carbon–oxygen bonds in a carboxylate anion given that the bond length of a C=O bond is 0.120 nm and a C—O bond is 0.134 nm. Justify your answer.
- Compare the bond angles and shapes around the two carbon atoms in the ethanoic acid ion.
- Compare the acidity of ethanoic acid with that of ethanol and phenol. Illustrate your answer by means of simple chemical tests.

Key ideas

- Carboxylic acids behave as acids by reacting with sodium hydroxide and with carbonates.
- Carboxylic acids are stronger acids than phenols because the anion is stabilised by the delocalisation of the negative charge over the carboxyl group.
- Evidence for the delocalisation of the carboxylate charge is seen in the intermediate bond length of the C—O bond.

Assignment 33.1: Lemon acid chemistry

Lemons are known for having a very sour taste which is due, in part, to the presence of citric acid. The structure of citric acid is:



Citric acid makes up between 5% and 6% by mass of the juice from a lemon (Figure 33.1). There are other acids in lemon juice, and these contribute to both the sour taste and the characteristic aroma of lemons. One of these acids is malic acid, see the structure later, but it is only about 0.25% by mass of lemon juice.

In the mid-1700s, James Lind discovered that citrus fruits such as lemons would help prevent sailors on long sea journeys from getting a disease called scurvy. This results in bleeding gums, tooth loss, jaundice and eventually death. The chemical present in lemons that helps to prevent scurvy is ascorbic acid. This is more commonly known as vitamin C. The World Health Organization recommendation is that adults need a minimum of 45 mg of vitamin C per day.

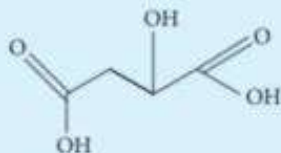
Tip

Citric acid is the common or trivial name. The systematic IUPAC name is 2-hydroxypropane-1,2,3-tricarboxylic acid. Understandably, citric acid is preferred for ease of use.



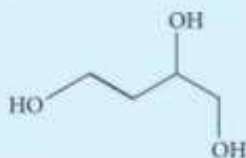
Figure 33.1 Lemons are a source of vitamin C.

- A1.** What is the molecular formula of citric acid?
- A2.** A sample of lemon juice contains about 5% by mass of citric acid. Calculate the concentration in mol dm^{-3} of citric acid in this sample of lemon juice. State any assumptions you make.
- A3.** One mole of citric acid reacts with how many moles of:
- sodium
 - sodium hydrogencarbonate?
- A4.** The structure of malic acid is:



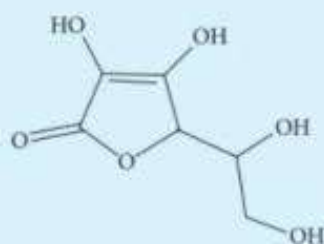
What is the systematic name of malic acid?

- A5.** Write the equation for the ionisation of one of the carboxylic acid groups of malic acid.
- A6.** Explain why it is very unlikely that this alcohol:



would be converted into malic acid by heating under reflux with acidified dichromate(VI).

A7. Vitamin C has the structure:



- Identify three functional groups in vitamin C.
- What functional group could be expected to be present in vitamin C from its name 'ascorbic acid'?
- How many lemons with a mass of 100 g each are required to provide the minimum daily amount of vitamin C for an adult? A lemon contains about 50 mg per 100 g.

33.5 Acidity of chlorine-substituted carboxylic acids

The statement has been made that 'carboxylic acids are weak acids' and this is generally true.

However, there can be exceptions to this general statement that carboxylic acids are weak acids. Note the difference in the pK_a value when the hydrogen atoms in the methyl group of ethanoic acid are substituted for chlorine atoms.

From the values in Table 33.1 we can conclude that trichloroethanoic acid is the strongest of the chlorine-substituted ethanoic acids.

Link

You may remember from Chapter 25 that pK_a is a measure of acid strength. The smaller the numerical value the stronger the acid.

Name	Displayed formula	pK_a
ethanoic acid		4.76
chloroethanoic acid		2.86
dichloroethanoic acid		1.29
trichloroethanoic acid		0.65

Table 33.1 pK_a values of ethanoic acid and chlorine-substituted ethanoic acids.

33 Carboxylic acids and derivatives

The strength of trichloroethanoic acid is clearly seen when the following pH values are compared:

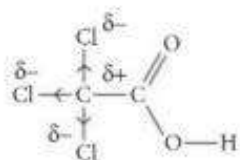
- A 0.10 mol dm^{-3} solution of a strong acid such as hydrochloric acid has a pH of 1.0.
- A 0.10 mol dm^{-3} solution of trichloroethanoic acid has a pH of 1.2.

Link

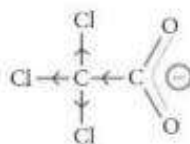
Chlorine is a highly electronegative element. A chlorine atom in a bond with carbon will attract more of the electron density of the covalent bond. See Chapter 3.

This means that we can view trichloroethanoic acid as a strong acid, namely one that is completely ionised, in contrast to ethanoic acid which is a weak acid and only partially ionised. We need to understand what is happening.

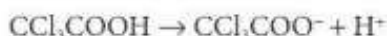
The result of the higher electronegativity of chlorine is that the C—Cl bond has a dipole with the chlorine being δ^- and the carbon δ^+ . In trichloroethanoic acid, this effect is multiplied three times:



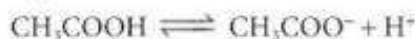
You can understand that this would have a stabilising effect on the carboxylate ion of trichloroethanoic acid. The negative charge on that ion will be pulled towards the three chlorine atoms and delocalised over the whole COO^- group which provides greater stability:



The effect of having three chlorine atoms in the molecule means that the ionisation of trichloroethanoic acid can be written as irreversible:

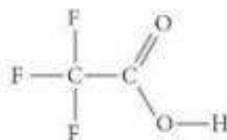


whereas, as previously written, the ionisation of ethanoic acid is reversible:



Worked example

Indicate on a structural diagram of trifluoroethanoic acid any relevant dipoles and describe any difference in acidity compared to trichloroethanoic acid.



Answer

The three fluorine atoms will be $-\delta$ and the central carbon atom will be $+\delta$. Fluorine is more electronegative than chlorine, so the dipoles will be larger than in trichloroethanoic acid. This means that the acidity of the fluorine-substituted ethanoic acid will be a stronger acid than the chlorine-substituted acid.

17. Suggest a $\text{p}K_a$ value for trifluoroethanoic acid. Justify your answer.
18. Write the equation for the reaction between chlorodifluoroethanoic acid and sodium carbonate.

Key ideas

- The presence of a chlorine atom in the alkyl group of a carboxylic acid increases the strength of the acid.
- The electronegative chlorine atom withdraws electron density from the carbon of the carboxyl group and helps to stabilise the anion.

33.6 Reactions of acyl chlorides

REACTION OF ACYL CHLORIDES WITH WATER

Acyl chlorides are often referred to as acid derivatives. The word 'derivative' in this context means that the compound has been made by replacing the $-OH$ of the carboxylic acid group with something else. The $-OH$ group is replaced by $-Cl$ to make acyl chlorides.

Most acyl chlorides are liquids at room temperature.

Name	Formula	Boiling point/ $^{\circ}C$
ethanoyl chloride	CH_3COCl	51
propanoyl chloride	CH_3CH_2COCl	80
butanoyl chloride	$CH_3CH_2CH_2COCl$	102
benzoyl chloride	C_6H_5COCl	197

Table 33.2 Boiling points of acyl chlorides.

The boiling points of the acyl chlorides in Table 33.2 are lower than the corresponding carboxylic acids. For example, ethanoic acid has a much higher boiling point of $118^{\circ}C$ compared to $51^{\circ}C$ for ethanoyl chloride. This is because of the intermolecular hydrogen bonding due to the $O-H$ bond in ethanoic acid. Weaker dipole-dipole forces are formed by the $C-Cl$ bond of ethanoyl chloride.

The simplest acyl chloride in Table 33.2 is ethanoyl chloride. This may raise the question 'what about methanoyl chloride, $HCOCl$?' This compound is unstable and decomposes to carbon monoxide and hydrogen chloride, so it can be ignored.

Worked example

State the intermolecular forces in propanoic acid and in propanoyl chloride. Use these to explain the difference in the boiling temperature of propanoic acid ($141^{\circ}C$) compared to that of propanoyl chloride ($80^{\circ}C$).

Answer

The intermolecular forces in propanoic acid are instantaneous dipole-induced dipole forces, permanent dipole-permanent dipole forces and hydrogen bonding.

The intermolecular forces in propanoyl chloride are instantaneous dipole-induced dipole forces and permanent dipole-permanent dipole forces.

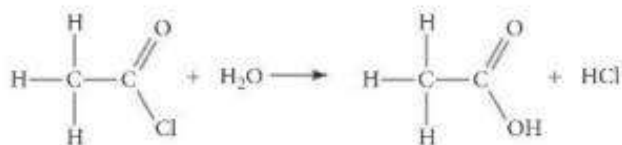
The instantaneous dipole-induced dipole forces in propanoyl chloride are stronger than those in propanoic acid because there are eight more electrons per molecule in propanoyl chloride. However, only propanoic acid has the stronger hydrogen bonds, which results in the much higher boiling temperature.

Link

Remember from Section 33.1 that acyl chlorides are made from the reaction of carboxylic acids with phosphorus chlorides or with sulfur dichloride oxide.

33 Carboxylic acids and derivatives

Acyl chlorides react with water rather than dissolving in it, so aqueous solutions of acyl chlorides do not exist. The reaction for ethanoyl chloride is:



Tip

The presence of hydrogen chloride gas can be identified by the formation of white smoke when it comes into contact with ammonia.

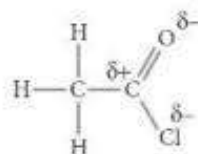
For this reason, acyl chlorides can be referred to as 'fuming' liquids because fumes of hydrogen chloride are given off when the liquid acyl chloride comes into contact with water (or even water vapour in the air). Acyl chlorides should always be used in a fume cupboard because immediately after the stopper is removed from the bottle fumes of hydrogen chloride are produced.

If a solution of an acyl chloride is required, then a dry organic solvent such as ether is used.

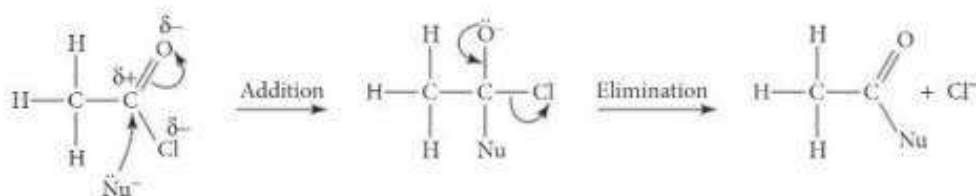
19. Draw a diagram to show the intermolecular forces that are present between two molecules of butanoic acid that are not present between two molecules of butanoyl chloride.
20. Explain why ethanoyl chloride is a liquid at room temperature.
21. Calculate the percentage of carbon by mass in benzoyl chloride.

REACTION BY AN ADDITION-ELIMINATION MECHANISM

The reaction with water is one example that shows acyl chlorides are much more reactive than carboxylic acids. The reason for this can be understood when the structure of acyl chlorides is considered. Note the structure of ethanoyl chloride:



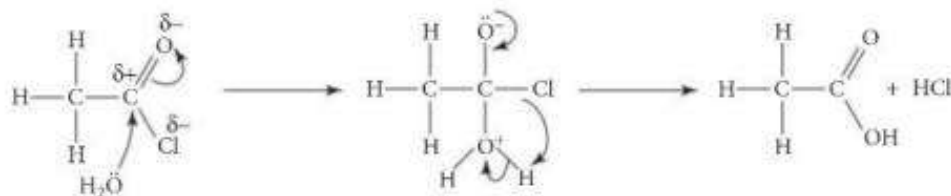
The carbon of the $-\text{COCl}$ group is bonded to two highly electronegative atoms. These atoms pull electron density from their bonds with the carbon atom more towards themselves. This results in the carbon atom having a significantly positive charge. This carbon atom is then easily attacked by a nucleophile and an addition-elimination reaction occurs. The general reaction with a nucleophile, Nu, is:



The initial attack by the nucleophile involves an addition reaction. The lone pair of electrons of the nucleophile form a covalent bond with the δ^+ carbon atom and at the same time the π bond of the $\text{C}=\text{O}$ group breaks. This produces an intermediate with a tetrahedral carbon atom with four substituents. This intermediate then quickly breaks down in an elimination reaction as the π bond reforms $\text{C}=\text{O}$ and the chloride ion is lost.

It is always best to understand and learn the principles of reaction mechanisms. In this way, you learn just a few reaction mechanism types rather than lots and lots of individual ones. The reaction of acyl chlorides in the following reactions to make esters and amides all occur via an addition-elimination reaction mechanism. However, for clarity let's focus

on the addition-elimination reaction involving ethanoyl chloride fuming in moist air or water. A simplified mechanism is:



The water molecule in moist air acts as a nucleophile with a lone pair of electrons on the oxygen atom being used to form a covalent bond between the oxygen and the $\delta+$ carbon atom. This results in the oxygen atom having three covalent bonds and so has a full positive charge. At the same time the π bond of the $\text{C}=\text{O}$ group breaks to leave an oxygen atom with a negative charge.

A lone pair of electrons on the oxygen atom with the negative charge reform the π bond and the $\text{C}=\text{O}$ group. At the same time the chlorine atom takes the two electrons from the $\text{C}-\text{Cl}$ covalent bond and uses them to form a covalent bond with a hydrogen atom attached to the other oxygen atom to form hydrogen chloride. The loss of the hydrogen atom means that the two electrons from the $\text{O}-\text{H}$ bond replace the lone pair of electrons on the oxygen atom from the initial water molecule.

Hydrogen chloride is formed rather than hydrochloric acid because the reaction of ethanoyl chloride with moist air is being considered. Water is limited and so the focus is its action as a nucleophile rather than a solvent.

If excess water is present the hydrogen chloride would dissolve and form hydrochloric acid.

Worked example

State and explain the changes in the bond angles around the carbon atoms in the reaction of ethanoyl chloride with water. Include the names of the relevant shapes from the bond angles.

Answer

In ethanoyl chloride, the bond angle around the methyl carbon is 109.5° because it is tetrahedrally bonded; and the bond angle around the carbonyl carbon is 120° because there are three areas of electron density. So the shape around the carbonyl carbon is trigonal planar. In the intermediate, both carbon atoms are tetrahedrally bonded, so the bond angles are 109.5° . In the carboxylic acid product the angles and shapes are the same as those for the acyl chloride.

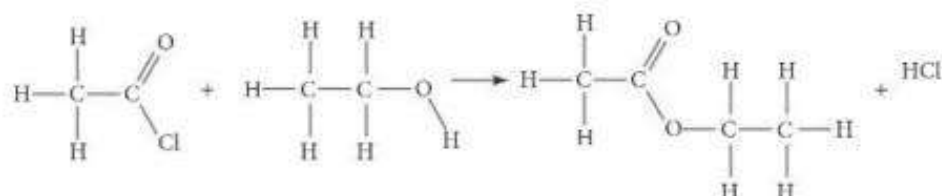
22. Define the terms 'nucleophile' and 'addition reaction'.
23. Which of the following is not a nucleophile?

A NH_3	C $\text{C}_2\text{H}_5\text{OH}$
B OH^-	D NH_4^+
24. Draw the reaction mechanism for the reaction of benzoyl chloride and water.

REACTIONS OF ACYL CHLORIDES TO PRODUCE ESTERS AND AMIDES

Acyl chlorides are highly reactive compounds that are very useful when synthesising other substances in a school or college laboratory on a small scale. The production of esters from their reaction with alcohols is one such use of acyl chlorides. The alcohols are acting as nucleophiles with the lone pair of electrons on the oxygen atom.

For example, ethyl ethanoate is an ester with a characteristic smell, similar to pear drops, and is used as a solvent in many glues and nail-polish removers. It can be made using the reaction between ethanoyl chloride and ethanol:

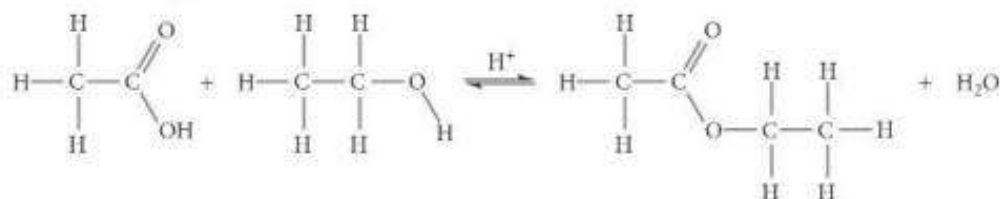


Worked example

Contrast the formation of the ester ethyl ethanoate using ethanoyl chloride with that using ethanoic acid.

Answer

The reaction using ethanoic acid is:



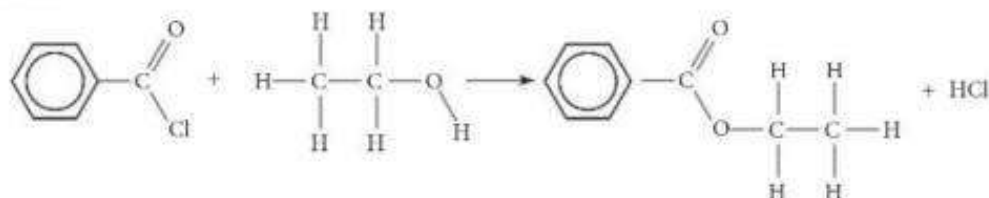
There are three differences to note:

- Carboxylic acids react with alcohols reversibly, but acyl chlorides react irreversibly.
- An acid catalyst is only required for esterification using carboxylic acids.
- Carboxylic acids produce a by-product of water, but acyl chlorides produce hydrogen chloride.

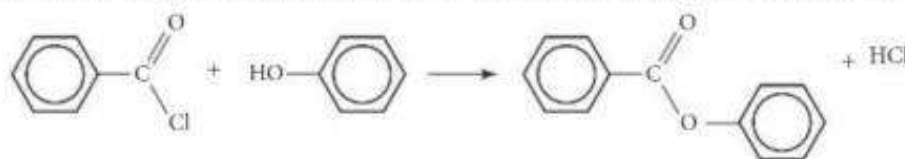
Tip

Remember that the part of the ester molecule joined to the oxygen by a single bond is the side chain, and the part with the $\text{C}=\text{O}$ is the stem.

Ethyl benzoate is an ester described as having a fruity aroma. It is commonly used in the production of perfumes. It is made using the reaction between benzoyl chloride and ethanol:



Phenyl benzoate is made using the reaction between benzoyl chloride and phenol:



All of these esterification reactions of acyl chlorides take place at room temperature because of the reactivity of these compounds.

Experimental skills 33.2: Preparation of aspirin

The carboxylic acid 2-hydroxybenzoic acid, commonly known as salicylic acid, is reacted with ethanoic anhydride by heating under reflux to form aspirin. The product is then purified by recrystallisation.

Note that ethanoyl chloride could be used instead of ethanoic anhydride. Because acyl chlorides are highly reactive, it is preferable on the scale used here to use the less reactive acid anhydride.

Apparatus

You will need the following apparatus:

- pear-shaped or round flask
- connector
- a condenser
- a heating mantle
- ethanoic anhydride
- 2-hydroxybenzoic acid (salicylic acid)
- a few anti-bumping granules
- concentrated phosphoric acid
- deionised water
- beaker
- glass rod
- Buchner funnel and filter paper
- Buchner flask
- vacuum pump
- ice bath
- kettle
- hot plate.

Technique

You will need access to a water supply, a sink and electricity.

The 2-hydroxybenzoic acid is placed in the flask along with a few anti-bumping granules. Add the ethanoic anhydride and a few drops of phosphoric acid. The flask is gently swirled and then heated under reflux for 10 minutes. After turning off the heat, deionised water is added down the condenser. This must be done carefully because the contents of the flask may boil. Once the mixture has cooled,

it is poured into a beaker of cold water to produce crude aspirin. You may need to stir with a glass rod in order to start precipitation.

Link

The apparatus for heating under reflux was used in Chapter 16 for the oxidation of primary alcohols.

The sample of crude aspirin is then filtered under reduced pressure using a Buchner funnel and flask. The solid aspirin sample is added to a minimum amount of hot water to dissolve the solid. The flask is cooled in an ice bath to re-precipitate the aspirin, which is then filtered under reduced pressure to give a purified sample of aspirin. The melting point of the aspirin can then be determined.

- P1. Why is the flask gently swirled during the initial reaction?
- P2. What can you infer about the reaction by the way water is added down the condenser?
- P3. During the recrystallisation step, why is the minimum amount of hot water used to dissolve the crude aspirin?
- P4. What type of impurities are removed by recrystallisation?
- P5. Draw a diagram of the apparatus for suction filtration.
- P6. How would hot filtration be carried out to remove a different type of impurity?
- P7. What needs to be done to the sample after filtration before the melting point is determined?

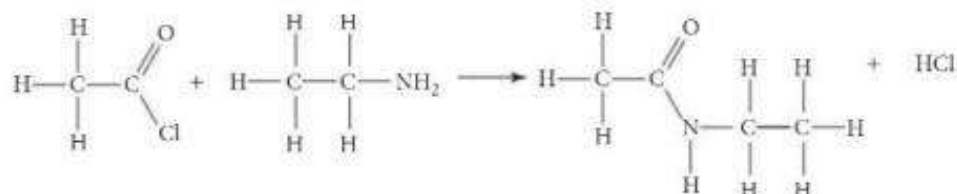
Ammonia has a nitrogen atom with a lone pair of electrons. This means that it can act as a nucleophile and react with acyl chlorides in a similar way to those previously considered. The reaction of ammonia with ethanoyl chloride at room temperature produces an amide, ethanamide:



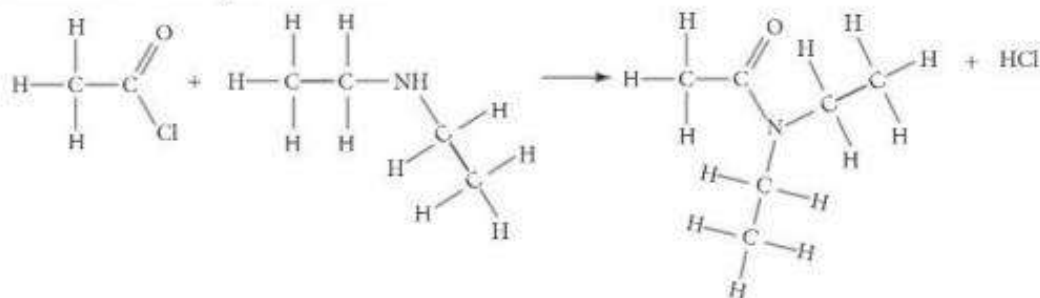
Amines also have a nitrogen atom with a lone pair of electrons, which means that they too can act as nucleophiles and react with acyl chlorides. The amides produced are referred to as substituted amides because one of the hydrogens attached to the nitrogen is replaced by the alkyl or aryl group of the amine.

33 Carboxylic acids and derivatives

Ethylamine is a primary amine and reacts with ethanoyl chloride at room temperature to make the **substituted amide** N-ethyl ethanamide:

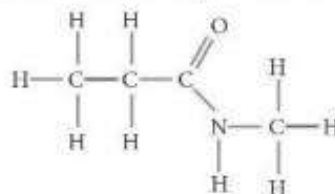


Diethylamine is a secondary amine. It reacts with ethanoyl chloride at room temperature to make *N,N*-diethylethanamide:



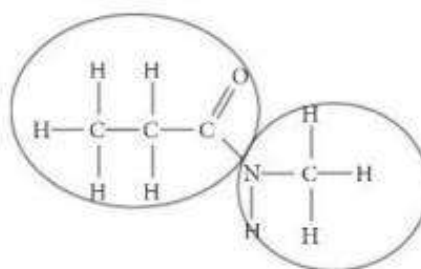
Worked example

Identify by name and formula the reactants required to form this amide:

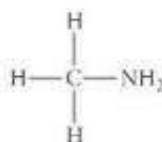


Answer

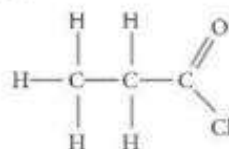
It can be helpful to start by drawing a circle or ring around the two different parts of the amide to help identify the two reactants that combined to form it. For example:



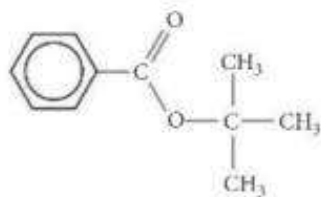
The nitrogen-containing reactant is a primary amine with one carbon and is called methylamine:



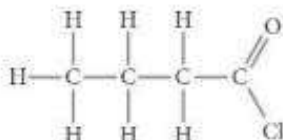
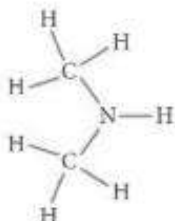
The carbonyl-containing reactant is an acyl chloride with three carbons in a straight chain. It is called propanoyl chloride:



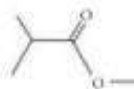
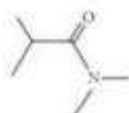
25. Identify by name and formula the reactants used to form the ester shown:



26. Identify by name and formula the amide formed from these reactants:



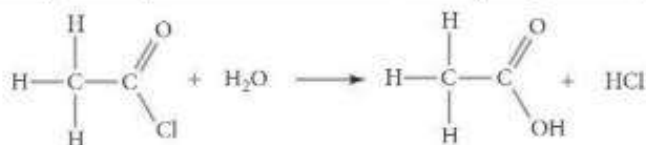
27. Deduce the displayed formula of the acyl chloride reagent that would be used to make both the ester and the amide shown.



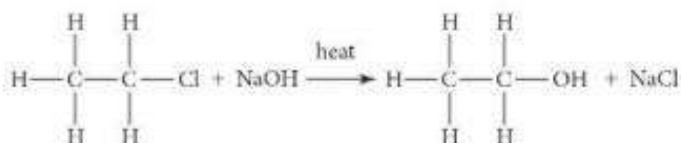
COMPARISON OF THE HYDROLYSIS OF ACYL CHLORIDES WITH THAT OF ALKYL CHLORIDES AND OF HALOGENOARENES

In this chapter you have considered the hydrolysis of acyl chlorides. In Chapter 15 you learned about the hydrolysis of halogenoalkanes such as chloroethane. In Chapter 30 you learned about the formation of halogenoarenes such as chlorobenzene. These three types of organic compound all contain C—Cl bonds that can be hydrolysed, or broken, and so provide an opportunity for comparison.

As you will remember from earlier in this chapter, acyl chlorides are hydrolysed very easily and react with water at room temperature. The δ^+ carbon atom of the C—Cl bond means that the weakly nucleophilic water molecule can hydrolyse the acyl chloride:



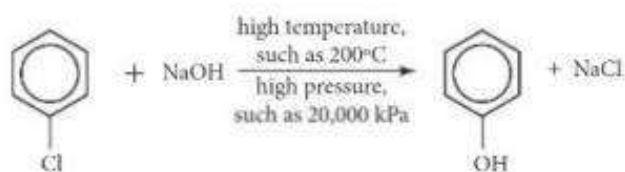
In Chapter 15 you studied the reactions of halogenoalkanes such as chloroethane, $\text{C}_2\text{H}_5\text{Cl}$. One of these was the reaction with aqueous sodium hydroxide, which requires heat under reflux:



Halogenoalkanes such as chloroethane require the stronger nucleophile, OH^- , and heat for hydrolysis to occur. The C—Cl bond of alkyl chlorides is stronger than in acyl chlorides and the carbon atom has a smaller δ^+ , so a more powerful nucleophile is required.

33 Carboxylic acids and derivatives

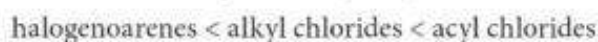
The hydrolysis of halogenoarenes such as chlorobenzene requires much harsher conditions:



The C—Cl bond in halogenoarenes is strengthened by the delocalisation of the lone pair of electrons in the chlorine atom with the ring of delocalised electrons. This extra electron delocalisation over the whole molecule increases the strength of the C—Cl bond. The result is that both high heat and high pressure are required for a reaction to occur.

An additional helpful way of remembering the difficulty of this reaction is to consider that the nucleophile is a negative hydroxide ion, OH^- , and this will be repelled by the area of high electron density of the arene ring.

In summary, the trend in increasing ease of hydrolysis is:

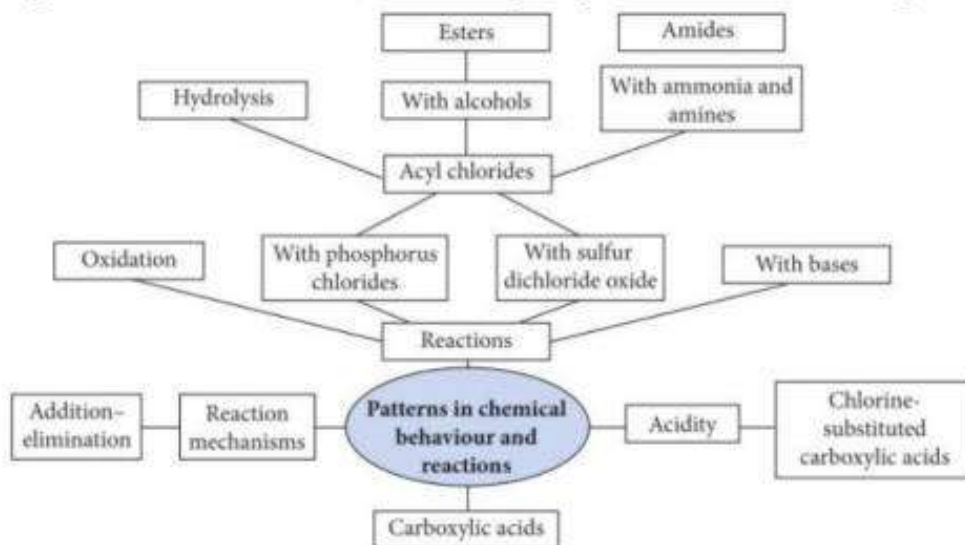


Key ideas

- Acyl chlorides are much more reactive than carboxylic acids.
- Acyl chlorides react by an addition–elimination mechanism.
- The addition–elimination mechanism starts with addition by a nucleophile and is then followed by the elimination of hydrogen chloride.
- Acyl chlorides react with alcohols to form esters.
- Acyl chlorides react with ammonia or with amines to form amides.
- Acyl chlorides are hydrolysed much more quickly than either halogenoalkanes or halogenoarenes.

CHAPTER OVERVIEW

Try copying this mini mind map and expanding it. Use your notes from other chapters to help you explore how the essential ideas, theories and principles can be linked further together.



WHAT YOU HAVE LEARNED

Now that you have finished this chapter you should be able to:

- recall a reaction by which benzoic acid can be produced
 - reaction of an alkylbenzene with hot alkaline KMnO_4 and then dilute acid, exemplified by methylbenzene

- describe the reaction of carboxylic acids with PCl_5 and heat, PCl_3 , or SOCl_2 to form acyl chlorides
- recognise that some carboxylic acids can be further oxidised:
 - the oxidation of methanoic acid, HCO_2H , with Fehling's reagent or Tollens' reagent or acidified KMnO_4 or acidified $\text{K}_2\text{Cr}_2\text{O}_7$ to carbon dioxide and water
 - the oxidation of ethanedioic acid, $\text{HO}_2\text{CCO}_2\text{H}$, with warm, acidified KMnO_4 to carbon dioxide
- describe and explain the relative acidities of carboxylic acids, phenols and alcohols
- describe and explain the relative acidities of chlorine-substituted carboxylic acids
- recall the reaction by which esters can be produced:
 - reaction of alcohols with acyl chlorides using the formation of ethyl ethanoate and phenyl benzoate as examples
- describe the following reactions of acyl chlorides:
 - the hydrolysis on addition of water at room temperature to give the carboxylic acid and HCl
 - the reaction with an alcohol at room temperature to give an ester and HCl
 - the reaction with phenol at room temperature to give an ester and HCl
 - the reaction with ammonia at room temperature to give an amide and HCl
 - the reaction with a primary or secondary amine at room temperature to give an amide and HCl
- describe the addition-elimination reaction in the reactions of acyl chlorides
- explain the relative ease of hydrolysis of acyl chlorides, alkyl chlorides and halogenoarenes.

CHAPTER REVIEW

- Name three different reagents that could be used to convert butanoic acid into butanoyl chloride.
 - Write equations for each of the three reactions. Include state symbols.
- State the two conditions required for sodium dichromate(VI) to oxidise butan-1-ol to butanoic acid.
- Describe how the oxidation of methanol is different to the oxidation of ethanol.
- Which two of the following reagents will oxidise methanoic acid?

A alcoholic ammonia	C Fehling's reagent
B Brady's reagent	D Tollens' reagent.
 - Describe the positive observation made with your chosen reagents.
- Identify the reagent, by name and formula, that will oxidise the compound HOCCOOH to carbon dioxide and water.
- Explain how sodium carbonate could be used to prove that phenol is a weaker acid than ethanoic acid.
- What is the evidence for the delocalisation of the charge on the carboxylate ion?
- The $\text{p}K_a$ of chloroethanoic acid (CH_2ClCOOH) is 2.86. Give one change that could be made to the structure of the acid to make it a stronger acid. Justify your choice and estimate the new $\text{p}K_a$ value.
- State what happens if you try to make methanoyl chloride.
- Describe what happens if you try to make an aqueous solution of ethanoyl chloride.
- Acyl chlorides are said to react by means of an addition-elimination mechanism. Explain this mechanism using the reaction of propanoyl chloride and water.
- Write the equation for the formation of the ester methyl ethanoate.
- Deduce the reactants required to produce *N,N*-dipropylbutanamide. Draw their displayed formulae.